

The hitchhiker's guide to **DATA WRANGLING**

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Based on slides by:
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✧ Grew up in Tuscany, Italy 🇮🇹

✧ Moved to London 🇬🇧 in 2012

✧ BSc in Biological Science, Royal Veterinary College

✧ MSc in Bioinformatics and Systems Biology, Birkbeck University

✧ Moved to The Netherlands 🇳🇱 in 2020 (🦠)

✧ PhD - *almost*

✧ **Keywords:** Semantic Table Interpretation, LLMs





Data wrangling

Data wrangling

what is it?

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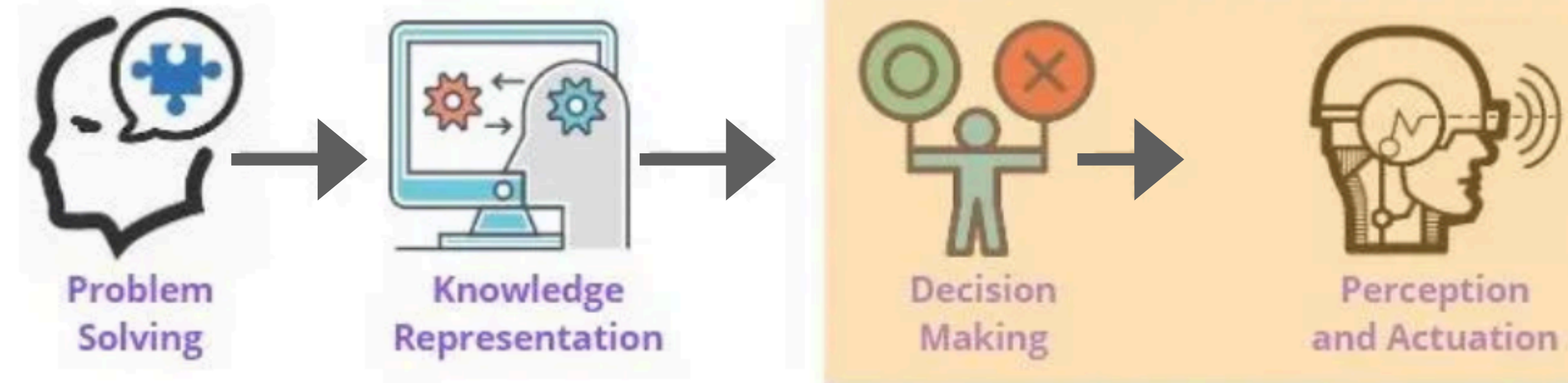
Data wrangling is the process of transforming and mapping data from one "raw" data form into another format

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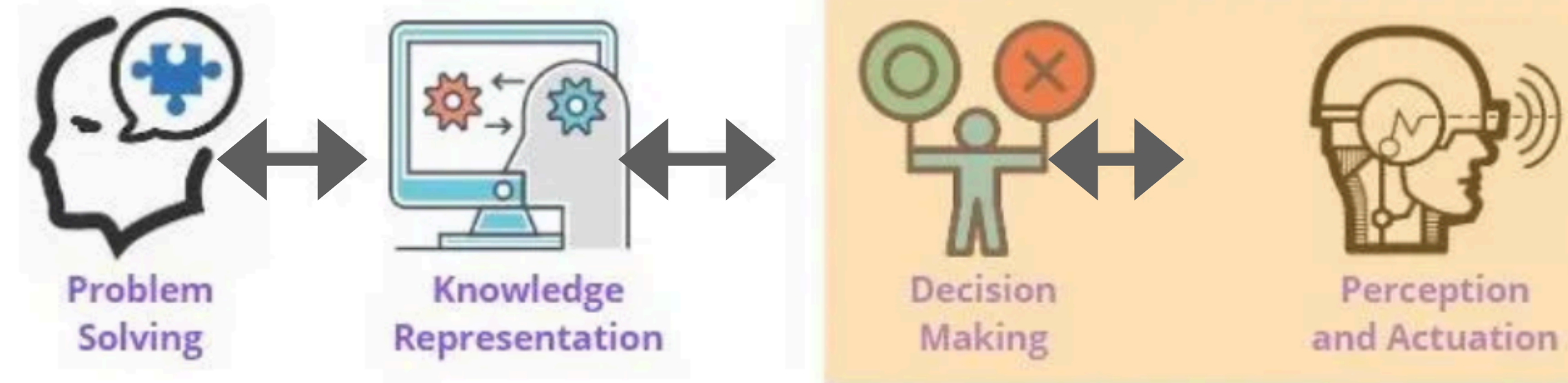
Data wrangling is the process of transforming and mapping data from one "raw" data form into another format with the intent of making it more appropriate and valuable ...

The goal of data wrangling is to assure quality and useful data.

Artificial Intelligence



Artificial Intelligence



Artificial Intelligence



Problem
Solving



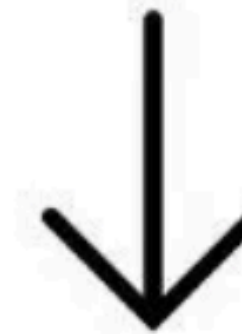
Knowledge
Representation



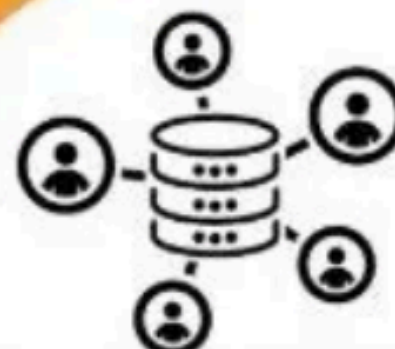
Decision
Making



Perception
and Actuation



These tasks
require
Data Science



Collect



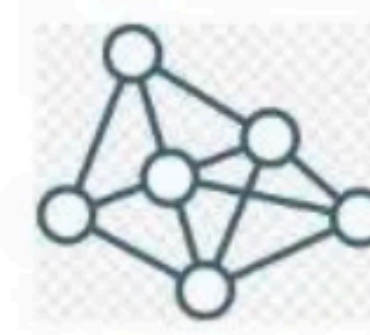
Store



Process



Describe



Model

Data Science

Artificial Intelligence



These tasks
require
Data Science



Data Science





DATABASE



DATABASE

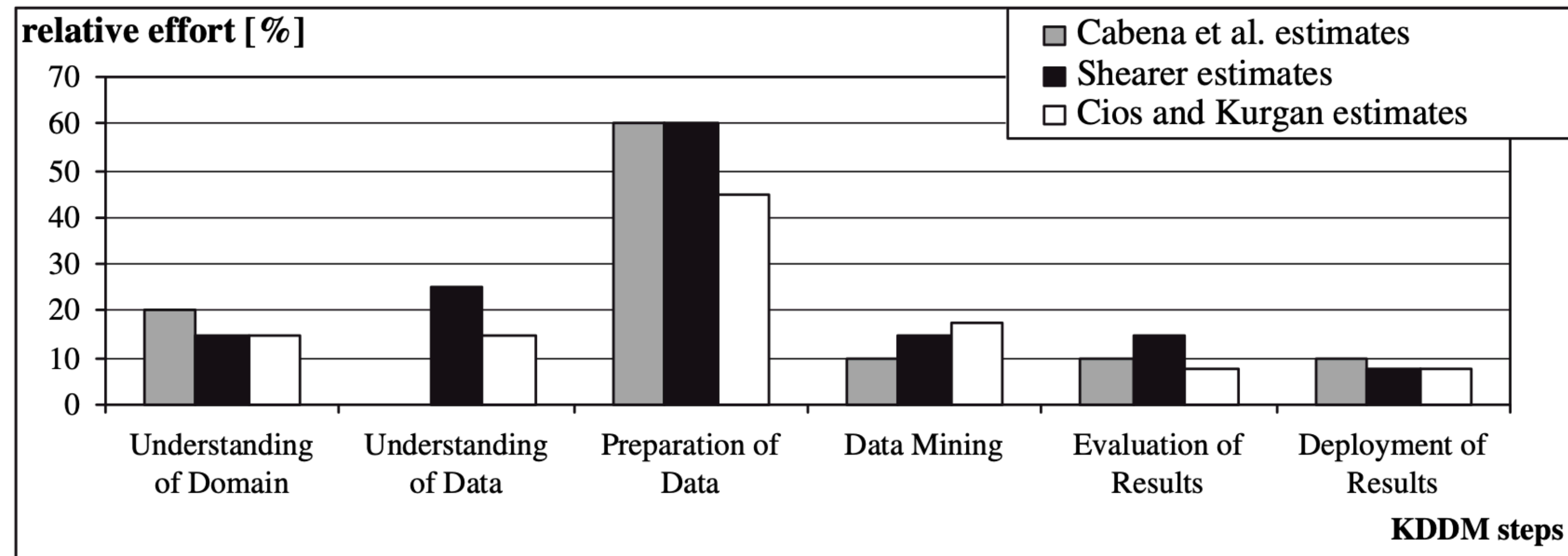


Figure 2.4. Relative effort spent on specific steps of the KD process. Source: Pal. N.R., Jain, L.C., (Eds.) 2005. Advanced Techniques in Knowledge Discovery and Data Mining, Springer Verlag.

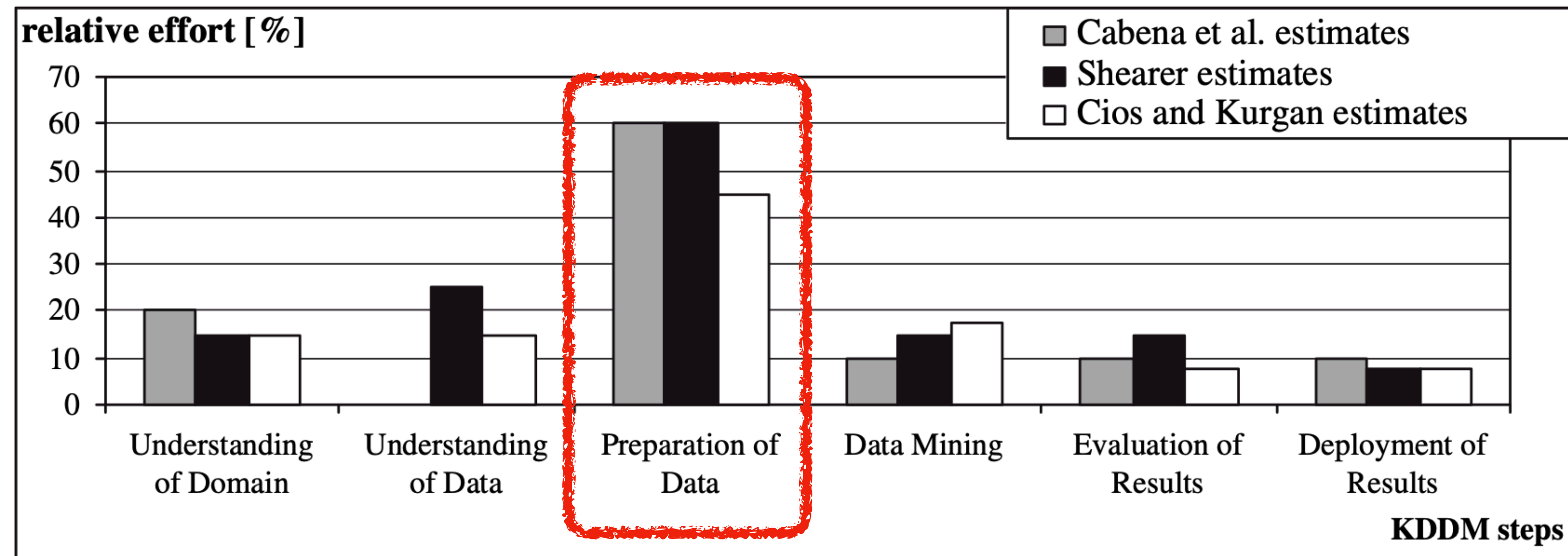


Figure 2.4. Relative effort spent on specific steps of the KD process. Source: Pal. N.R., Jain, L.C., (Eds.) 2005. Advanced Techniques in Knowledge Discovery and Data Mining, Springer Verlag.

Enterprise Data Analysis and Visualization: An Interview Study

Sean Kandel, Andreas Paepcke, Joseph M. Hellerstein, and Jeffrey Heer

Abstract—Organizations rely on data analysts to model customer engagement, streamline operations, improve production, inform business decisions, and combat fraud. Though numerous analysis and visualization tools have been built to improve the scale and efficiency at which analysts can work, there has been little research on how analysis takes place within the social and organizational context of companies. To better understand the enterprise analysts’ ecosystem, we conducted semi-structured interviews with 35 data analysts from 25 organizations across a variety of sectors, including healthcare, retail, marketing and finance. Based on our interview data, we characterize the process of industrial data analysis and document how organizational features of an enterprise impact it. We describe recurring pain points, outstanding challenges, and barriers to adoption for visual analytic tools. Finally, we discuss design implications and opportunities for visual analysis research.

Index Terms—Data, analysis, visualization, enterprise.

1 INTRODUCTION

Organizations gather increasingly large and complex data sets each year. These organizations rely on data analysis to model customer engagement, streamline operations, improve production, inform sales and business decisions, and combat fraud. Within organizations, an increasing number of individuals—with varied titles such as “business analyst”, “data analyst” and “data scientist”—perform such analyses. These analysts constitute an important and rapidly growing user population for analysis and visualization tools.

Enterprise analysts perform their work within the context of a larger organization. Analysts often work as a part of an analysis team or business unit. Little research has observed how existing infrastructure, available data and tools, and administrative and social conventions within an organization impact the analysis process within the enterprise. Understanding how these issues shape analytic workflows can inform the design of future tools.

To better understand the day-to-day practices of enterprise analysts, we conducted semi-structured interviews with 35 analysts from sectors including healthcare, retail, finance, and social networking. We asked analysts to walk us through the typical tasks they perform, the tools they use, the challenges they encounter, and the organizational context in which analysis takes place.

In this paper, we present the results and analysis of these interviews. We find that our respondents are well-described by **three archetypes** that differ in terms of skill set and typical workflows. We find that these archetypes vary widely in programming proficiency, reliance on information technology (IT) staff and diversity of tasks, and vary less in statistical proficiency. We then discuss how organizational features of an enterprise, such as the relationship between analysts and IT staff or the diversity of data sources, affect the analysis process. We also describe how collaboration takes place between analysts. We find that analysts seldom share resources such as scripts and intermediate data products. In response, we consider possible impediments to sharing and collaboration.

Next we characterize the analysis process described to us by our respondents. Our model includes **five high-level tasks**: *discovery*, *wrangling*, *profiling*, *modeling* and *reporting*. We find that discov-

ery and wrangling, often the most tedious and time-consuming aspects of an analysis, are underserved by existing visualization and analysis tools. We discuss recurring pain points within each task as well as difficulties in managing workflows across these tasks. Example pain points include integrating data from distributed data sources, visualizing data at scale and operationalizing workflows. These challenges are typically more acute within large organizations with a diverse and distributed set of data sources.

We conclude with a discussion of future trends and the implications of our interviews for future visualization and analysis tools. We argue that future visual analysis tools should leverage existing infrastructures for data processing to enable scale and limit data migration. One avenue for achieving better interoperability is through systems that specify analysis or data processing operations in a high-level language, enabling retargeting across tools or platforms. We also note that the current lack of reusable workflows could be improved via less intrusive methods for recording data provenance.

2 RELATED WORK

Many researchers have studied analysts and their processes within intelligence agencies [5, 18, 24, 25, 30]. This work characterizes intelligence analysts’ process, discusses challenges within the process, and describes collaboration among analysts. Although there is much overlap in the high-level analytic process of intelligence and enterprise analysts, these analysts often work on different types of data with different analytic goals and therefore perform different low-level tasks. For example, enterprise analysts more often perform analysis on structured data than on documents and emails.

Others have characterized tasks and challenges within the analysis process [1, 21, 26, 34, 35]. Amar et al. [1] propose a set of low-level analysis tasks based on the activities of students in an Information Visualization class. Their taxonomy largely includes tasks subsumed by our *profile* and *model* tasks and does not address the other tasks we have identified. Russell et al. [34] characterize high-level sensemaking activities necessary for analysis. We instead identify specific tasks performed by enterprise analysts. Sedlmair et al. [35] discuss difficulties evaluating visualization tools in large corporations, including acquiring and integrating data. We discuss common challenges within these subtasks. Kwon and Fisher [26] discuss challenges novices encounter when using visual analytic tools. In contrast, our subjects are largely expert users of their tools.

Fink et al. [9] performed an ethnographic study of cyber-security analysts. They find that visual analytic tools in this domain have limited interoperability with other tools, lack support for performing ad hoc transformations of data, and typically do not scale to the necessary volume and diversity of data. We find similar issues across multiple domains.

Several researchers have articulated the importance of capturing provenance to manage analytic workflows [2, 11, 12, 15]. Such sys-

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- Jeffrey Heer is with Stanford University, e-mail: jheer@cs.stanford.edu.

Manuscript received 31 March 2012; accepted 1 August 2012; posted online 14 October 2012; mailed on 5 October 2012.
For information on obtaining reprints of this article, please send e-mail to: tvcg@computer.org.

Interview study of 35 analysts

25 companies

Healthcare

Marketing

Media

Finance

Insurance

Various titles

Data Analyst

Data Scientist

Software Engineer

Consultant

CTO

Kandel, Sean, et al. "Enterprise data analysis and visualization: An interview study." IEEE transactions on visualization and computer graphics 18.12 (2012): 2917-2926.

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Most of the time I'm lucky if I get to do any 'analysis' at all...

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... once you transform the data ... the insights can be scarily obvious.”

“Once you play with the data you realize you made an **assumption that is completely wrong**. It’s really useful, it’s not just a waste of time, even though you may be banging your head.”

“Once you play with the data you realize you made an **assumption that is completely wrong**. It’s really useful, it’s not just a waste of time, even though you may be banging your head.”

“In practice it tends not to be just data prep, you are learning about the data at the same time, **you are learning about what assumptions you can make**.”



**Big Data
Borat**

@BigDataBorat



Following

In Data Science, 80% of time spent prepare data, 20% of time spent complain about need for prepare data.



When we DATA WRANGLE we are ...

... assessing, manipulating and transforming
raw data to make it fit for use

When we DATA WRANGLE we are ...

... assessing, manipulating and transforming
raw data to make it fit for use



... yeah but for what use?

THAT DEPENDS!





UNDERSTAND
WHAT YOU
HAVE



UNDERSTAND
WHAT YOU
HAVE



ASSESS THE
STRENGTHS &
WEAKNESSES





UNDERSTAND
WHAT YOU
HAVE



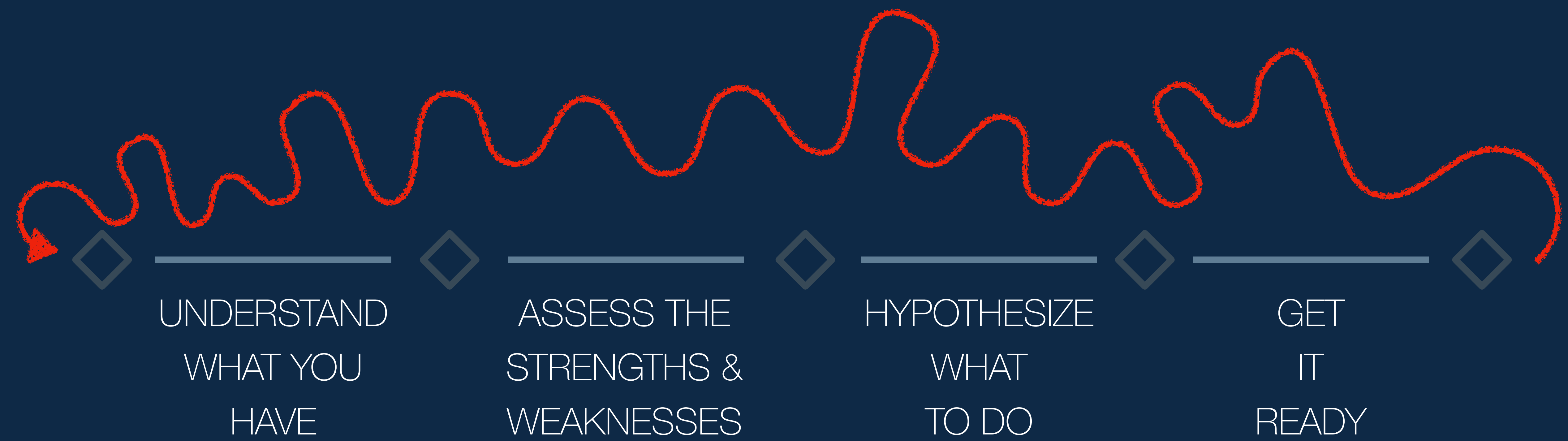
ASSESS THE
STRENGTHS &
WEAKNESSES

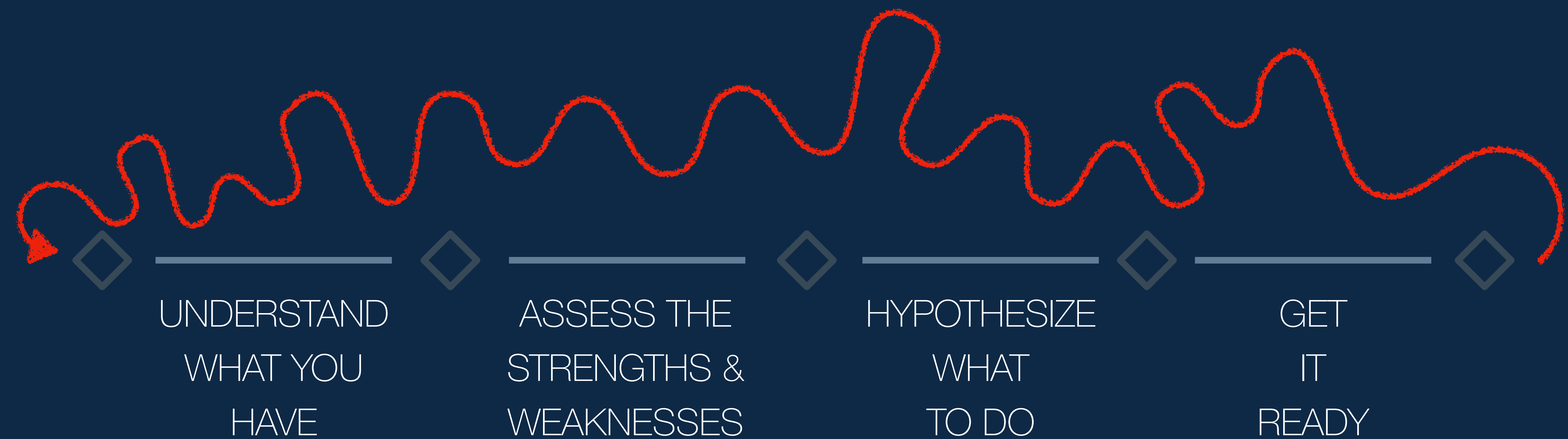


HYPOTHESIZE
WHAT
TO DO

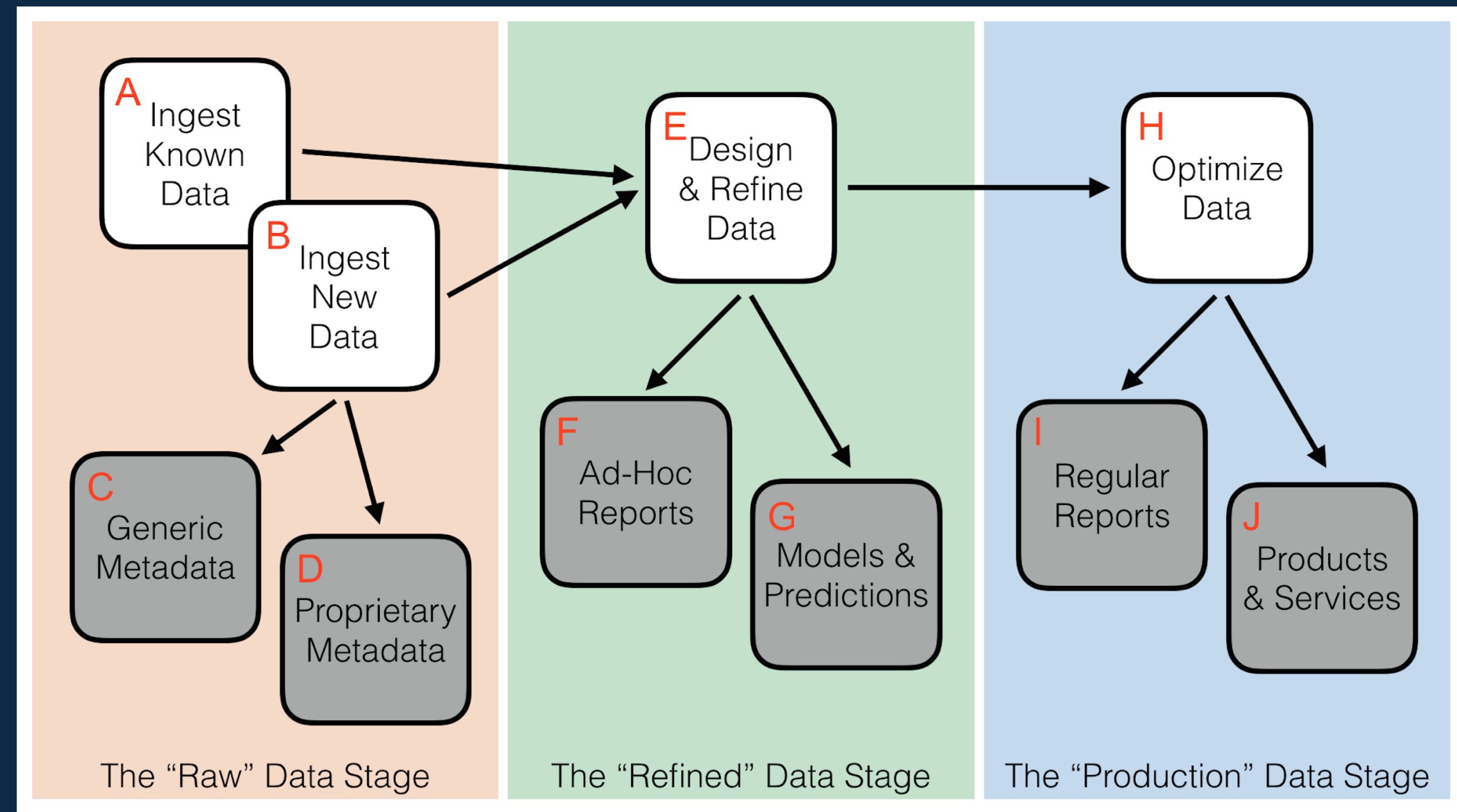


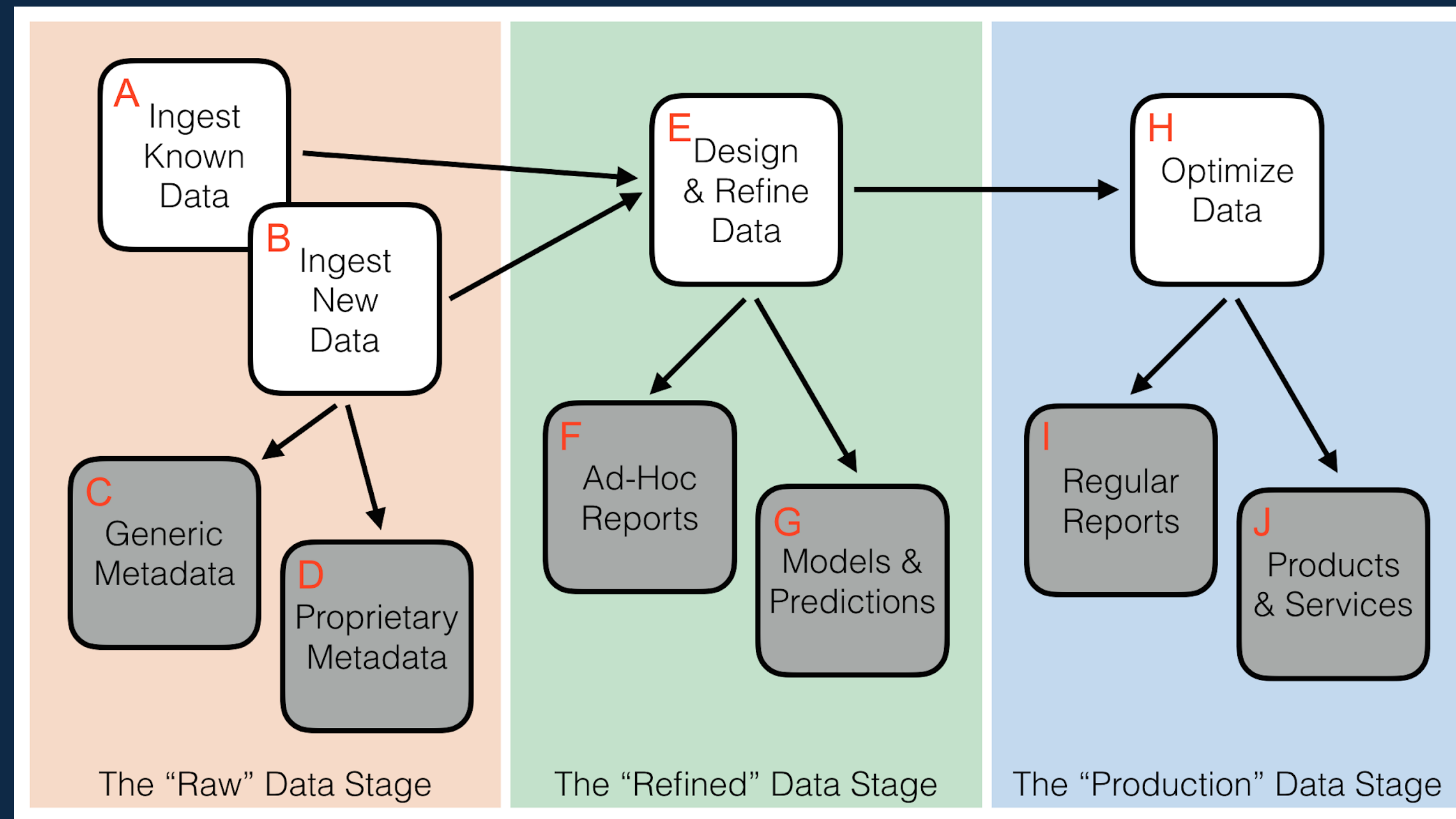






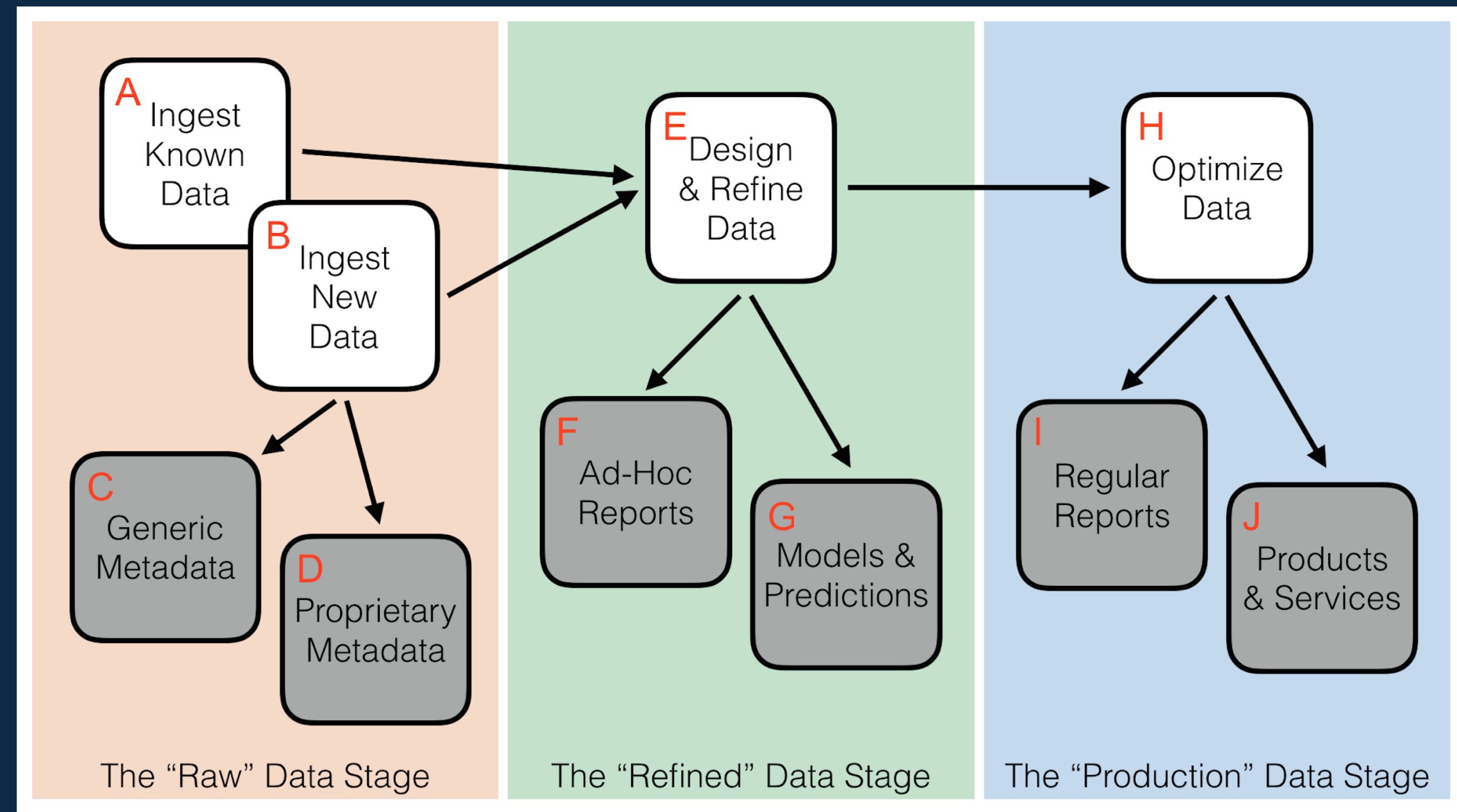
FIRST GET IT DONE, THEN GET IT GOOD





DATA INGESTION & DISCOVERY 'UNBOXING'

- exploratory *ad hoc* analysis
- individual wrangler

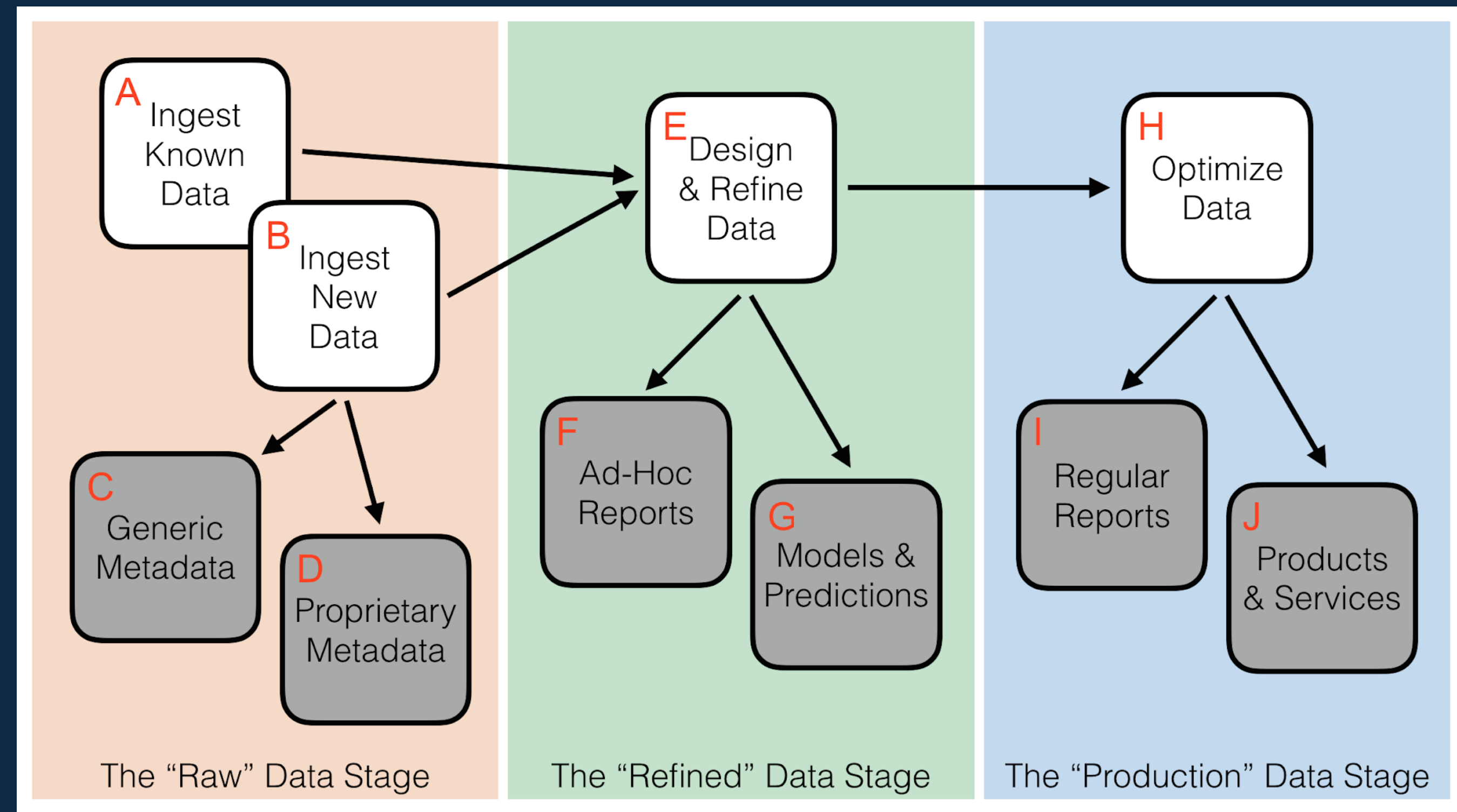


DATA INGESTION & DISCOVERY 'UNBOXING'

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CURATING DATA FOR REUSE

- databases, canonical models
- data curators, IT engineers ...



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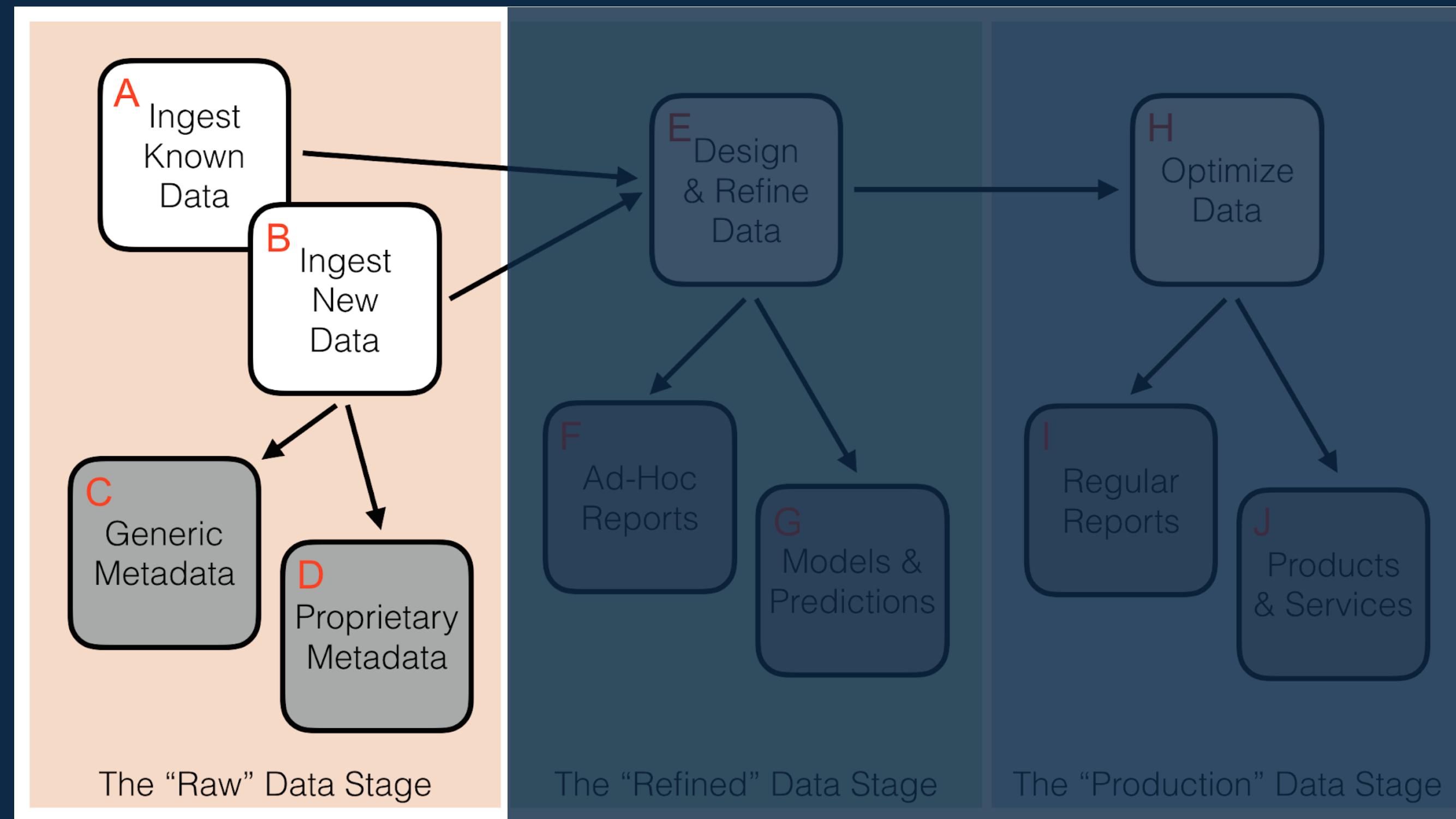
CURATING DATA FOR REUSE

- databases, canonical models
- data curators, IT engineers ...

FEEDBACKS AND WORKFLOWS

- use cases/recommenders
- SW/IT/ops engineers

TODAY!



DATA INGESTION & DISCOVERY 'UNBOXING'

- exploratory *ad hoc* analysis
- individual wrangler

RAW → REFINED

- ✱ 'unboxing'
- ✱ transformation to analytics-ready structure
- ✱ assessment/mitigation of quality issues



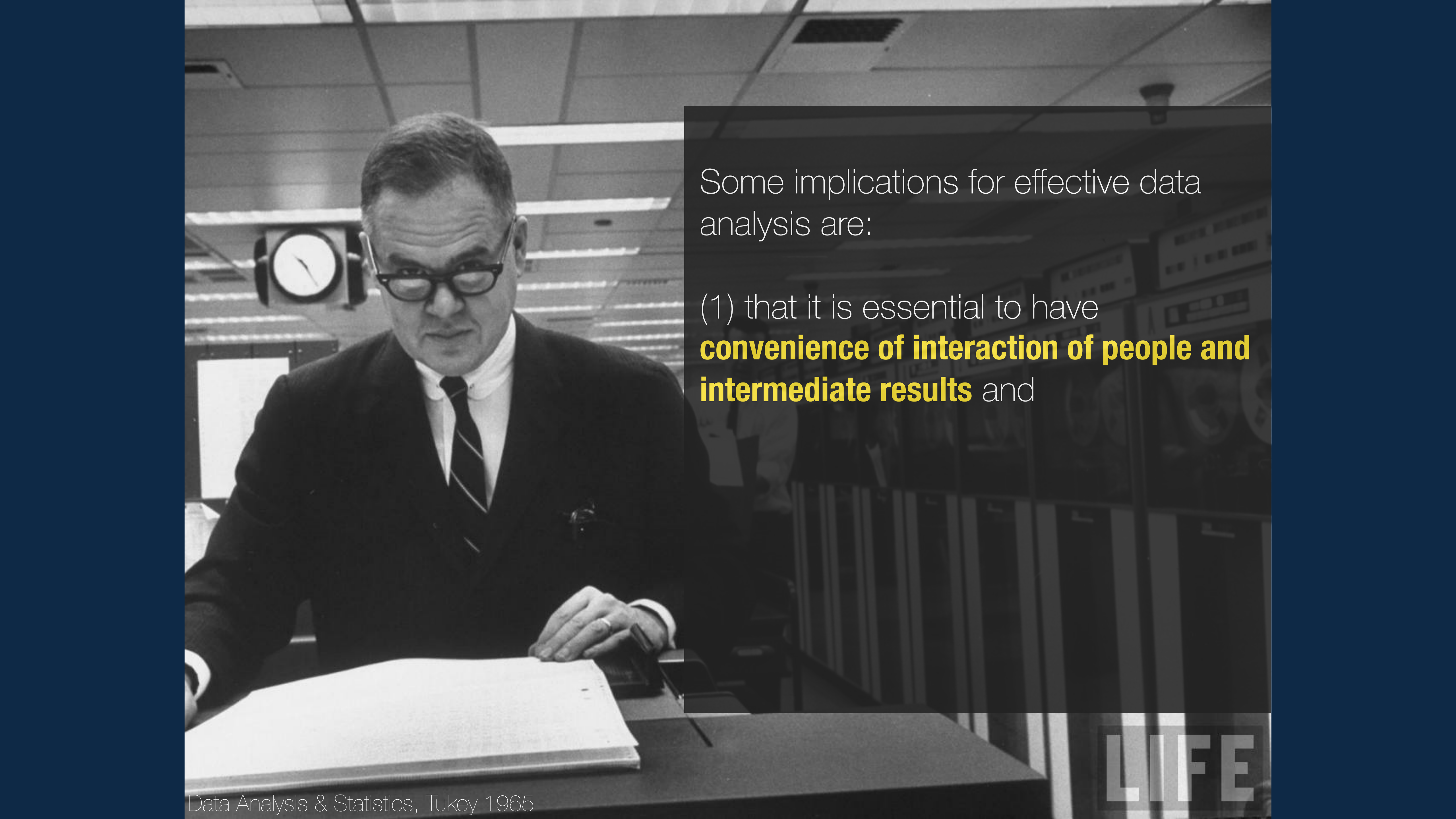
Dirty
Data





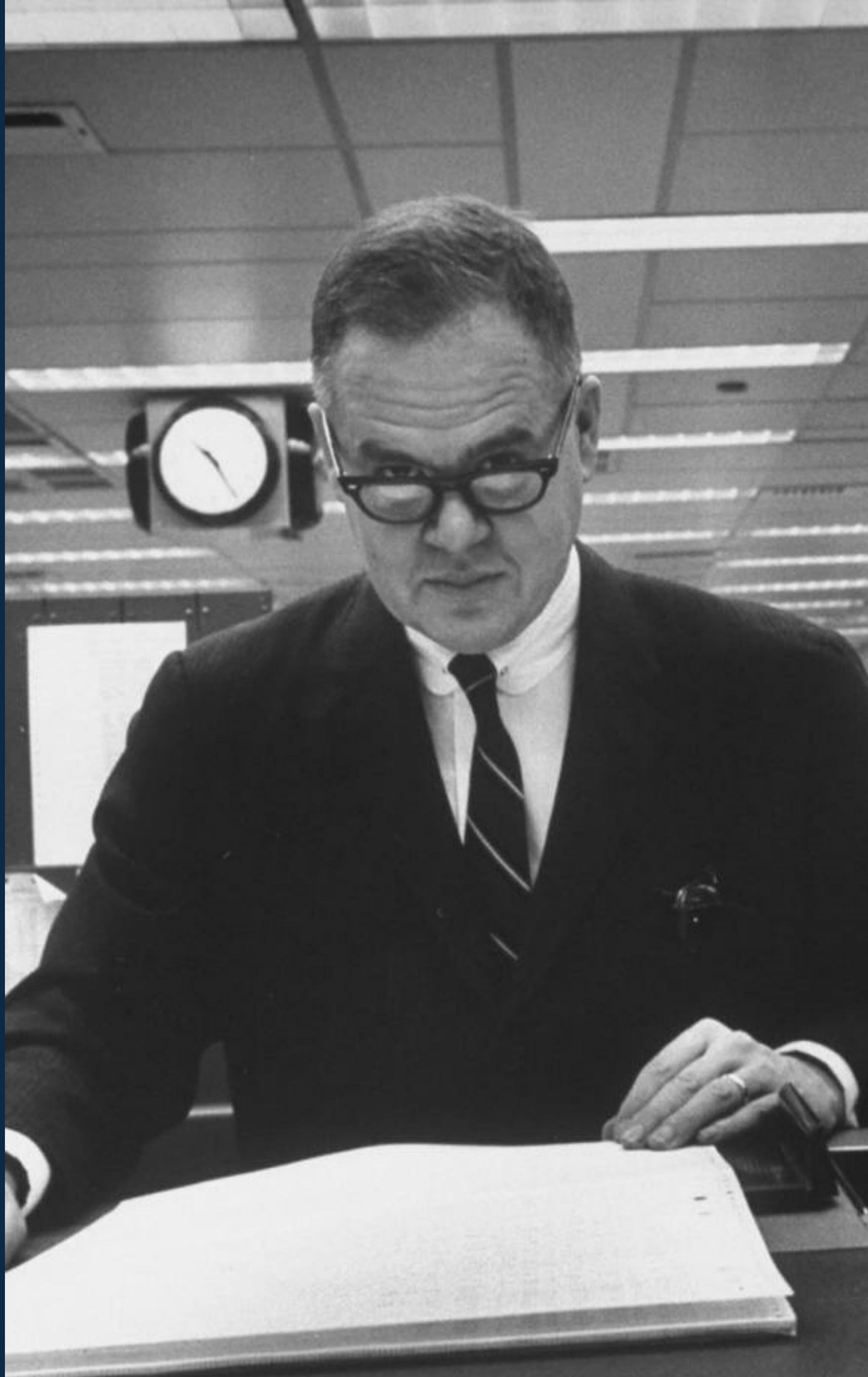
Data Analysis & Statistics, Tukey 1965

LIFE



Some implications for effective data analysis are:

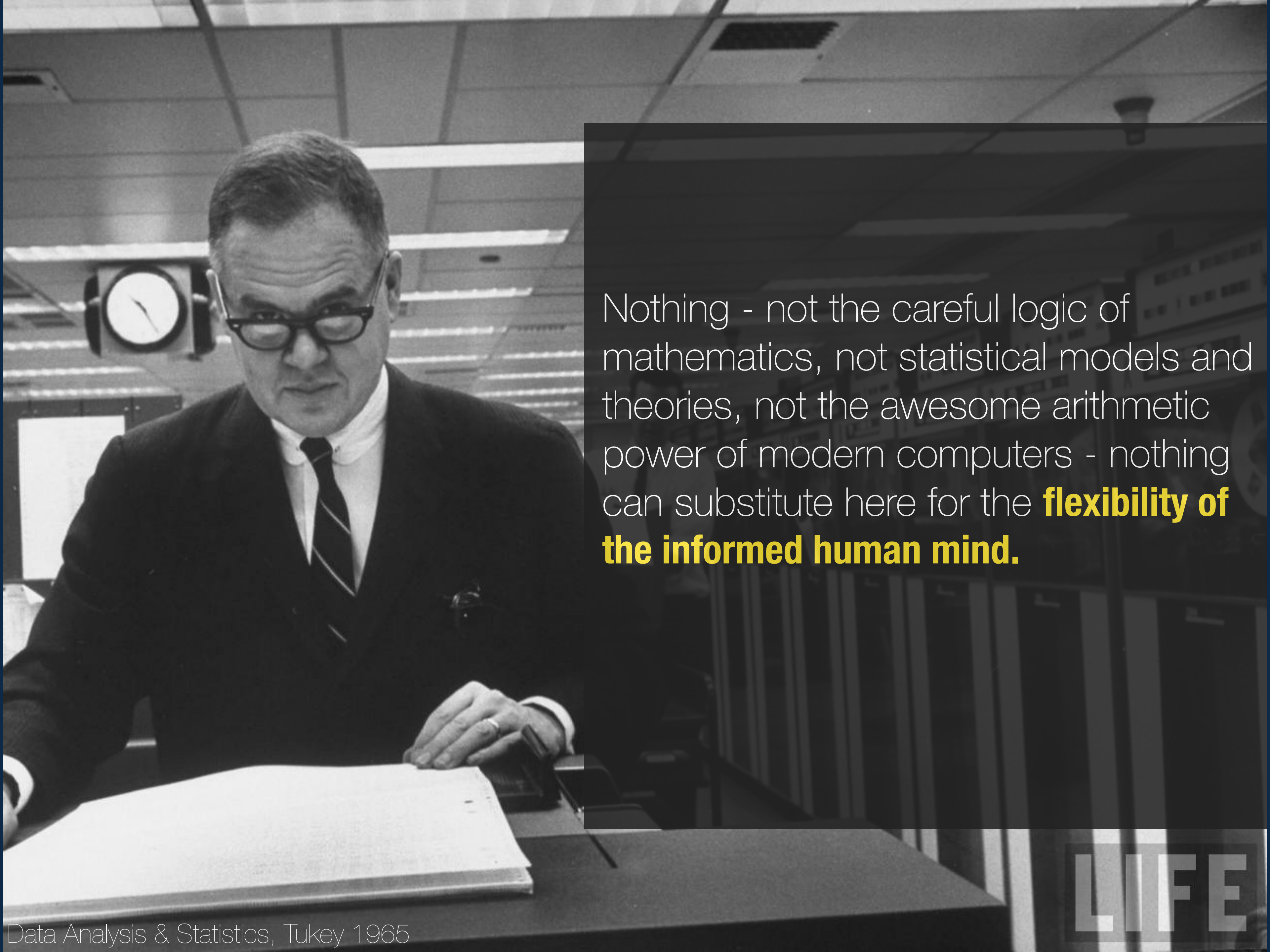
(1) that it is essential to have **convenience of interaction of people and intermediate results** and



Some implications for effective data analysis are:

(1) that it is essential to have **convenience of interaction of people and intermediate results** and

(2) that at all stages of data analysis, the **nature and detail of output**, both actual and potential, **need to be matched to the capabilities of the people** who use it and want it.



Nothing - not the careful logic of mathematics, not statistical models and theories, not the awesome arithmetic power of modern computers - nothing can substitute here for the **flexibility of the informed human mind.**



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LIFE

**WHAT DO
I HAVE
HERE?**



**WHAT DO I
WANT TO DO
WITH IT?**

**WHAT DO
I HAVE
HERE?**



**WHAT DO I
WANT TO DO
WITH IT?**

CONTEXT and USER-DRIVEN

ITERATION of WRANGLING and ANALYSIS

The hitchhiker's guide to WRANGLING

1. STRUCTURE: the 'shape' of a data file

The hitchhiker's guide to WRANGLING

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2. GRANULARITY: how fine/coarse is each datum

The hitchhiker's guide to WRANGLING

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The hitchhiker's guide to WRANGLING

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5. SCOPE: how (in)complete is the data

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SUBJECTIVE and **CONTEXT-DEPENDENT** features

(DATA) SCIENCE is a **HUMAN PROCESS**

RECTANGULAR DATA

natural solution for data analysis:
easy to filter and access

RECTANGULAR DATA



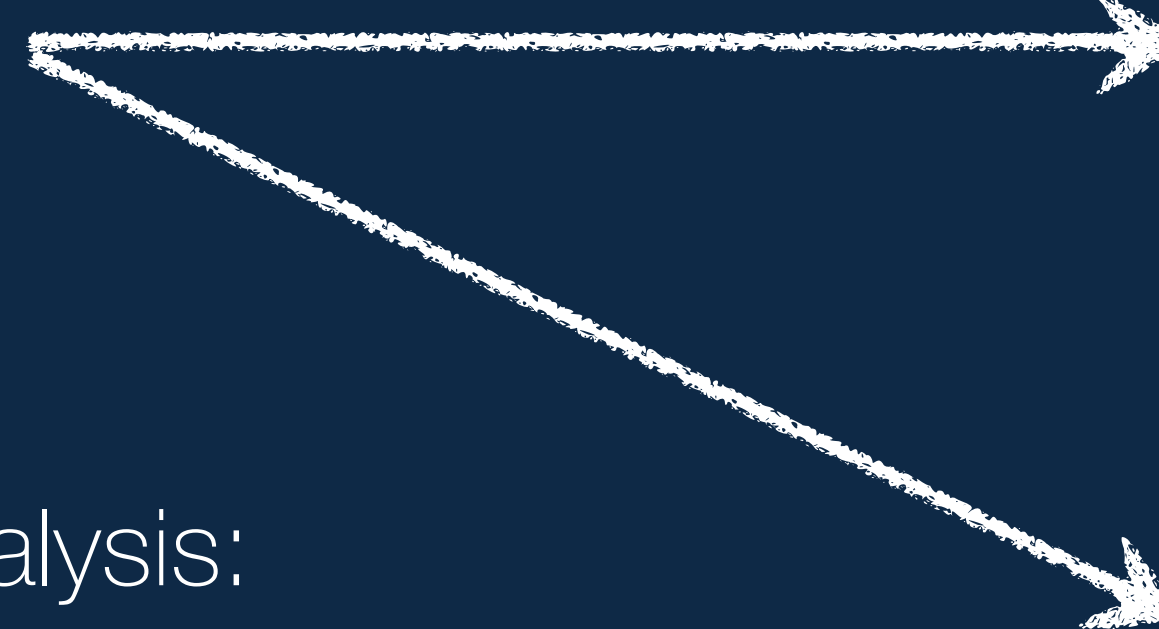
relations (e.g. tables, dataframes)
- manipulation with relational algebra

natural solution for data analysis:

easy to filter and access

RECTANGULAR DATA

natural solution for data analysis:
easy to filter and access

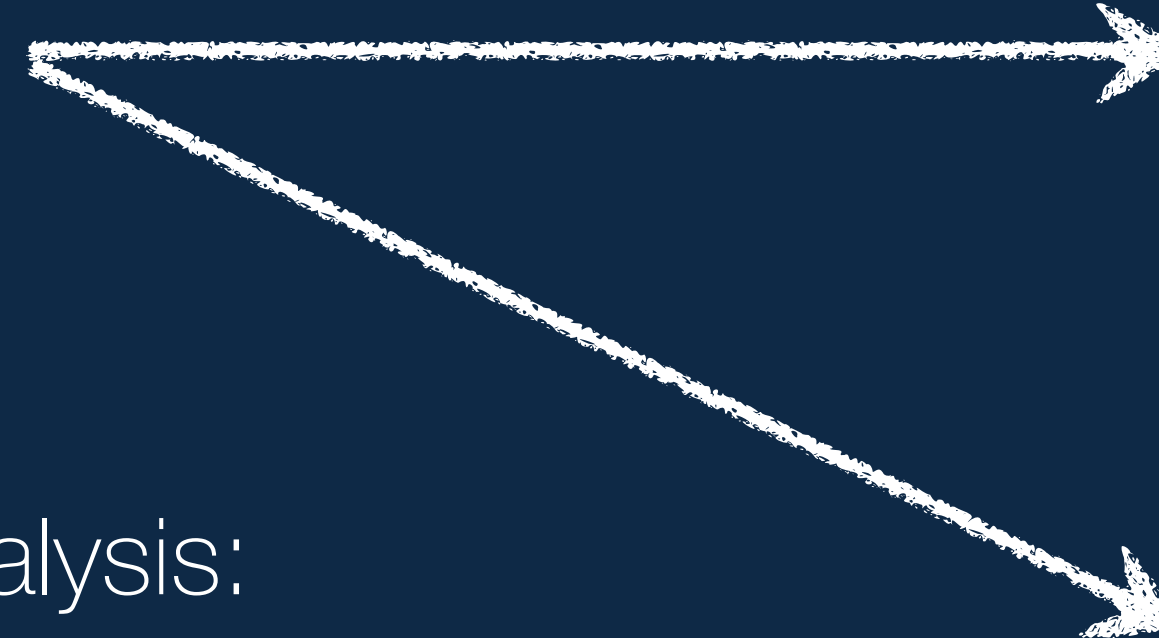


relations (e.g. tables, dataframes)
- manipulation with relational algebra

matrices
- manipulation with linear algebra

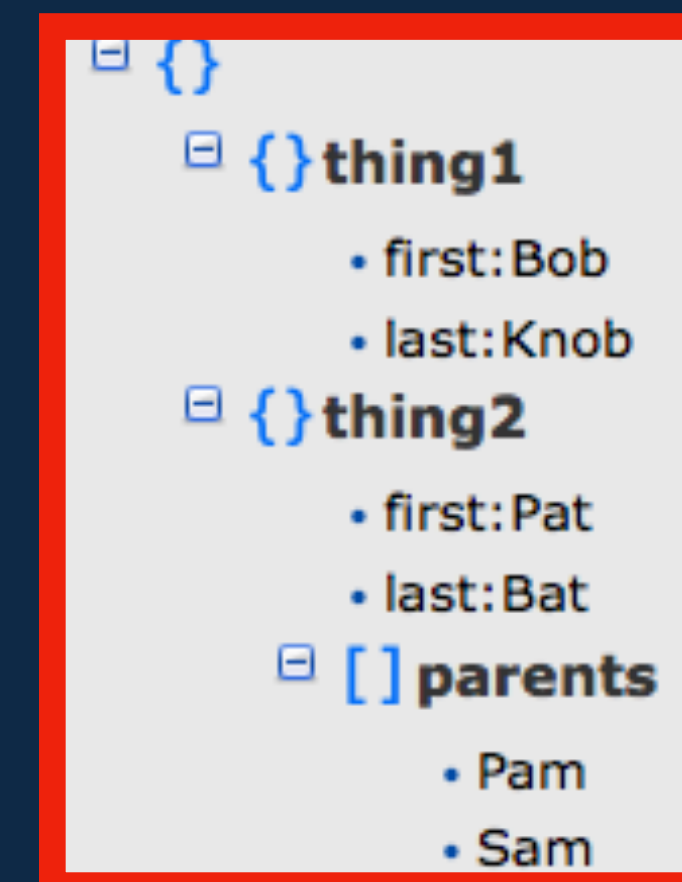
RECTANGULAR DATA

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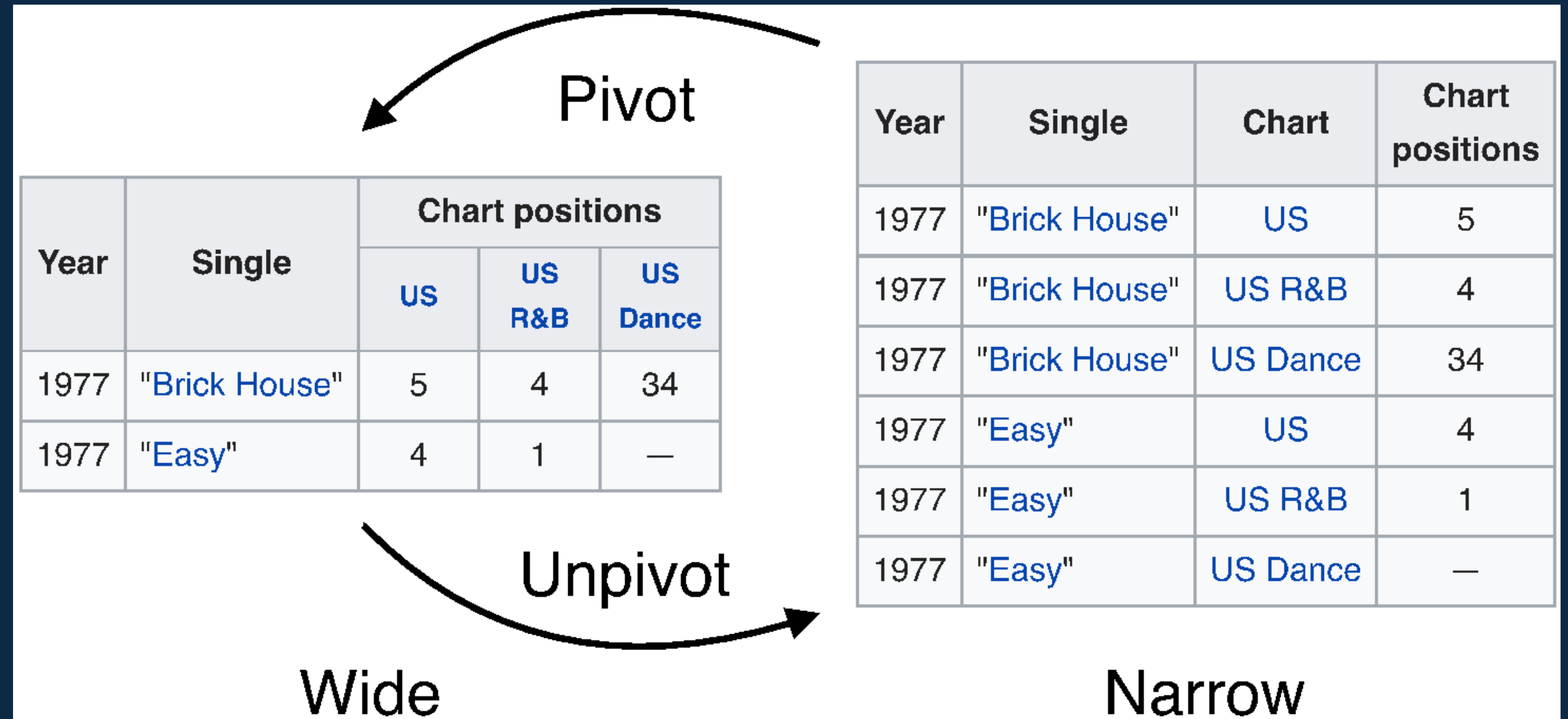


relations (e.g. tables, dataframes)
- manipulation with relational algebra

matrices
- manipulation with linear algebra



RECTANGULAR DATA



HANDS ON PRACTICE



 <http://bennokruit.nl/iai/titanic.csv>

File > Save as ...

 <https://hub.compute.vu.nl/>

hub2 or hub3 > Launch

Environment Selector > Python3 > Sign in with VU credentials



 Upload File > titanic.csv

Double click > Open With ... > Editor

Open With ... > CSV Viewer

Open With ... > Spreadsheet Editor

WHAT ARE THE DIFFERENCES?

Launcher > Terminal

File metadata —> ls -lh <file>

File content —> cat <file>
head <file>
tail <file>
less <file>

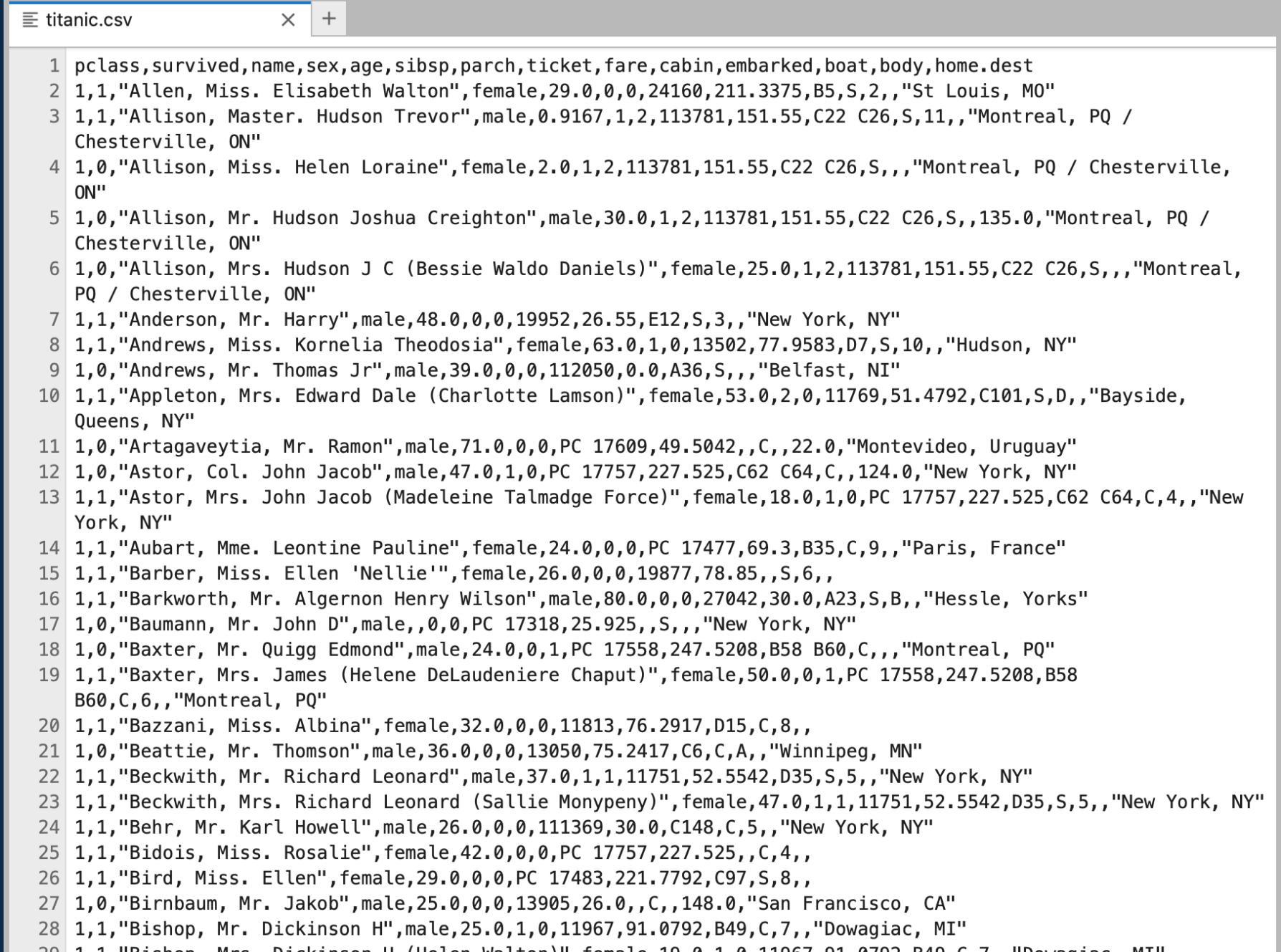
WHAT ARE THE DIFFERENCES?

STRUCTURE: A CHECKLIST

1. COARSE STRUCTURE

high-level organisation or overall shape

- Is the data structured as a collection of **records**?
- How are the individual records delimited?
- How are the record **fields** delimited?
- Do all records contain the same fields?
- How to access the same fields across records?
e.g. by position, name



```
titanic.csv
1 pclass,survived,name,sex,age,sibsp,parch,ticket,fare,cabin,embarked,boat,body,home.dest
2 1,1,"Allen, Miss. Elisabeth Walton",female,29.0,0,0,24160,211.3375,B5,S,2,,,"St Louis, MO"
3 1,1,"Allison, Master. Hudson Trevor",male,0.9167,1,2,113781,151.55,C22 C26,S,11,,,"Montreal, PQ /
  Chesterville, ON"
4 1,0,"Allison, Miss. Helen Loraine",female,2.0,1,2,113781,151.55,C22 C26,S,,,,"Montreal, PQ / Chesterville,
  ON"
5 1,0,"Allison, Mr. Hudson Joshua Creighton",male,30.0,1,2,113781,151.55,C22 C26,S,,135.0,"Montreal, PQ /
  Chesterville, ON"
6 1,0,"Allison, Mrs. Hudson J C (Bessie Waldo Daniels)",female,25.0,1,2,113781,151.55,C22 C26,S,,,,"Montreal,
  PQ / Chesterville, ON"
7 1,1,"Anderson, Mr. Harry",male,48.0,0,0,19952,26.55,E12,S,3,,,"New York, NY"
8 1,1,"Andrews, Miss. Kornelia Theodosia",female,63.0,1,0,13502,77.9583,D7,S,10,,,"Hudson, NY"
9 1,0,"Andrews, Mr. Thomas Jr",male,39.0,0,0,112050,0.0,A36,S,,,,"Belfast, NI"
10 1,1,"Appleton, Mrs. Edward Dale (Charlotte Lamson)",female,53.0,2,0,11769,51.4792,C101,S,D,,,"Bayside,
  Queens, NY"
11 1,0,"Artagaveytia, Mr. Ramon",male,71.0,0,0,PC 17609,49.5042,,C,22.0,"Montevideo, Uruguay"
12 1,0,"Astor, Col. John Jacob",male,47.0,1,0,PC 17757,227.525,C62 C64,C,,124.0,"New York, NY"
13 1,1,"Astor, Mrs. John Jacob (Madeleine Talmadge Force)",female,18.0,1,0,PC 17757,227.525,C62 C64,C,4,,,"New
  York, NY"
14 1,1,"Aubart, Mme. Leontine Pauline",female,24.0,0,0,PC 17477,69.3,B35,C,9,,,"Paris, France"
15 1,1,"Barber, Miss. Ellen 'Nellie'",female,26.0,0,0,19877,78.85,,S,6,,,"
16 1,1,"Barkworth, Mr. Algernon Henry Wilson",male,80.0,0,0,27042,30.0,A23,S,B,,,"Hessle, Yorks"
17 1,0,"Baumann, Mr. John D",male,,0,0,PC 17318,25.925,,S,,,,"New York, NY"
18 1,0,"Baxter, Mr. Quigg Edmond",male,24.0,0,1,PC 17558,247.5208,B58 B60,C,,,,"Montreal, PQ"
19 1,1,"Baxter, Mrs. James (Helene DeLaunier Chaput)",female,50.0,0,1,PC 17558,247.5208,B58
  B60,C,6,,,"Montreal, PQ"
20 1,1,"Bazzani, Miss. Albina",female,32.0,0,0,11813,76.2917,D15,C,8,,,"
21 1,0,"Beattie, Mr. Thomson",male,36.0,0,0,13050,75.2417,C6,C,A,,,"Winnipeg, MN"
22 1,1,"Beckwith, Mr. Richard Leonard",male,37.0,1,1,11751,52.5542,D35,S,5,,,"New York, NY"
23 1,1,"Beckwith, Mrs. Richard Leonard (Sallie Monypeny)",female,47.0,1,1,11751,52.5542,D35,S,5,,,"New York, NY"
24 1,1,"Behr, Mr. Karl Howell",male,26.0,0,0,111369,30.0,C148,C,5,,,"New York, NY"
25 1,1,"Bidois, Miss. Rosalie",female,42.0,0,0,PC 17757,227.525,,C,4,,,"
26 1,1,"Bird, Miss. Ellen",female,29.0,0,0,PC 17483,221.7792,C97,S,8,,,"
27 1,0,"Birnbaum, Mr. Jakob",male,25.0,0,0,13905,26.0,,C,148.0,"San Francisco, CA"
28 1,1,"Bishop, Mr. Dickinson H",male,25.0,1,0,11967,91.0792,B49,C,7,,,"Dowagiac, MI"
29 1,1,"Bishop, Mrs. Dickinson H (Helen Walton)",female,19.0,1,0,11967,91.0792,B49,C,7,,,"Dowagiac, MI"
```

2. ARE THE RECORDS NESTED?

- NO - one atomic (singular) value per field
- YES - collection of 0-to-many values in a field

```
titanic.csv
1 pclass,survived,name,sex,age,sibsp,parch,ticket,fare,cabin,embarked,boat,body,home.dest
2 1,1,"Allen, Miss. Elisabeth Walton",female,29.0,0,0,24160,211.3375,B5,S,2,,,"St Louis, MO"
3 1,1,"Allison, Master. Hudson Trevor",male,0.9167,1,2,113781,151.55,C22 C26,S,11,,,"Montreal, PQ /
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16 1,1,"Barkworth, Mr. Algernon Henry Wilson",male,80.0,0,0,27042,30.0,A23,S,B,,,"Hessle, Yorks"
17 1,0,"Baumann, Mr. John D",male,,0,0,PC 17318,25.925,,S,,,,"New York, NY"
18 1,0,"Baxter, Mr. Quigg Edmond",male,24.0,0,1,PC 17558,247.5208,B58 B60,C,,,,"Montreal, PQ"
19 1,1,"Baxter, Mrs. James (Helene DeLaunierie Chaput)",female,50.0,0,1,PC 17558,247.5208,B58
20 1,1,"Bazzani, Miss. Albina",female,32.0,0,0,11813,76.2917,D15,C,8,,,"
21 1,0,"Beattie, Mr. Thomson",male,36.0,0,0,13050,75.2417,C6,C,A,,,"Winnipeg, MN"
22 1,1,"Beckwith, Mr. Richard Leonard",male,37.0,1,1,11751,52.5542,D35,S,5,,,"New York, NY"
23 1,1,"Beckwith, Mrs. Richard Leonard (Sallie Monypeny)",female,47.0,1,1,11751,52.5542,D35,S,5,,,"New York, NY"
24 1,1,"Behr, Mr. Karl Howell",male,26.0,0,0,111369,30.0,C148,C,5,,,"New York, NY"
25 1,1,"Bidois, Miss. Rosalie",female,42.0,0,0,PC 17757,227.525,,C,4,,,"
26 1,1,"Bird, Miss. Ellen",female,29.0,0,0,PC 17483,221.7792,C97,S,8,,,"
27 1,0,"Birnbaum, Mr. Jakob",male,25.0,0,0,13905,26.0,,C,,148.0,"San Francisco, CA"
28 1,1,"Bishop, Mr. Dickinson H",male,25.0,1,0,11967,91.0792,B49,C,7,,,"Dowagiac, MI"
29 1,1,"Bishop, Mrs. Dickinson H (Helen Walton)",female,19.0,1,0,11967,91.0792,B49,C,7,,,"Dowagiac, MI"
```


2. ARE THE RECORDS NESTED?

- **NO** - one atomic (singular) value per field
- **YES** - collection of 0-to-many values in a field

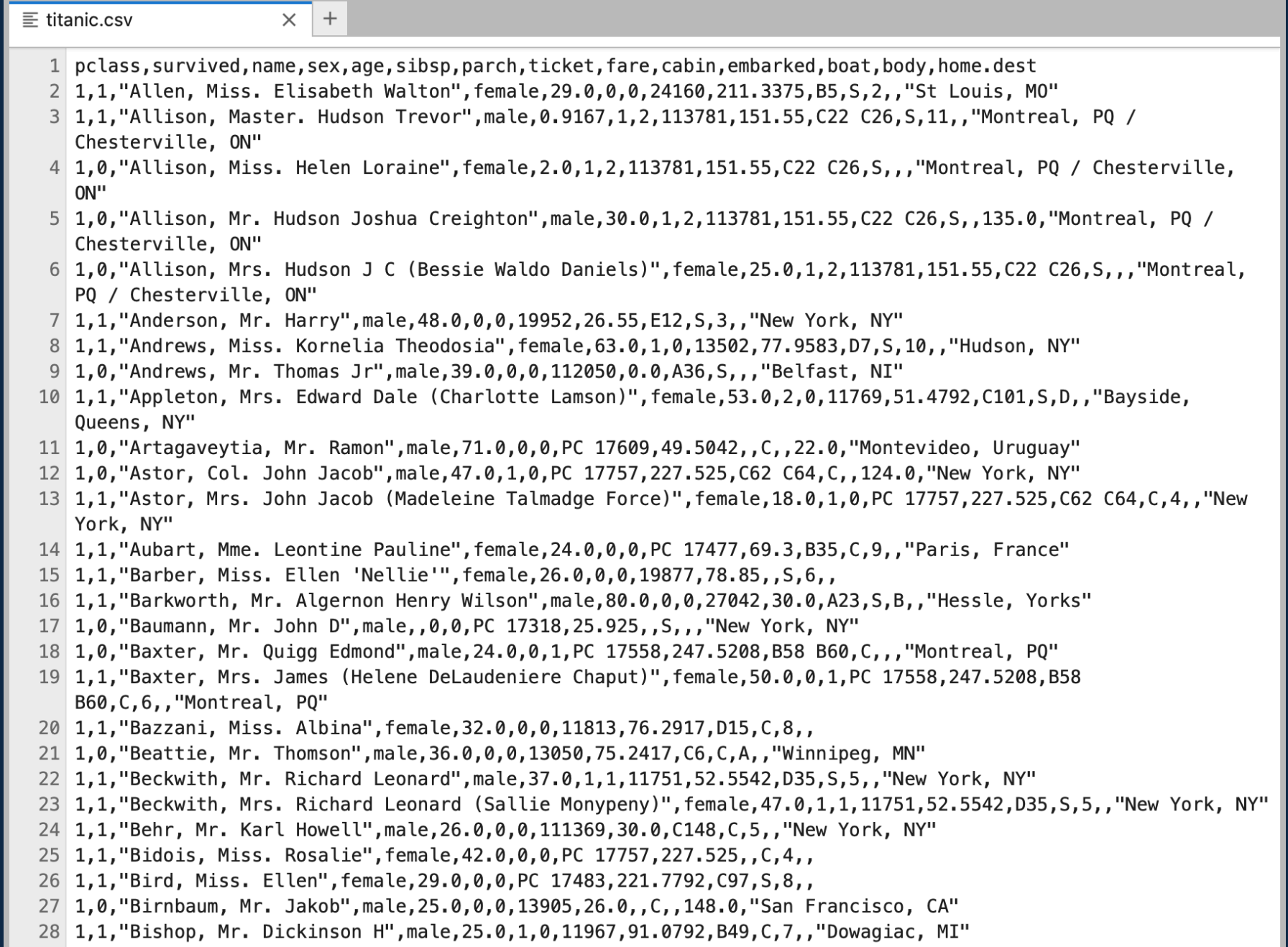
3. ENCODING

- How are values encoded? Strings? Binary? Code?
- How complex are the individual values?

PRIMITIVE - numbers, short strings?

UNSTRUCTURED - natural language text, audio, video?

- Can you decode it?



```
1 pclass,survived,name,sex,age,sibsp,parch,ticket,fare,cabin,embarked,boat,body,home.dest
2 1,1,"Allen, Miss. Elisabeth Walton",female,29.0,0,0,24160,211.3375,B5,S,2,,,"St Louis, MO"
3 1,1,"Allison, Master. Hudson Trevor",male,0.9167,1,2,113781,151.55,C22 C26,S,11,,,"Montreal, PQ /
  Chesterville, ON"
4 1,0,"Allison, Miss. Helen Loraine",female,2.0,1,2,113781,151.55,C22 C26,S,,,,"Montreal, PQ / Chesterville,
  ON"
5 1,0,"Allison, Mr. Hudson Joshua Creighton",male,30.0,1,2,113781,151.55,C22 C26,S,,135.0,"Montreal, PQ /
  Chesterville, ON"
6 1,0,"Allison, Mrs. Hudson J C (Bessie Waldo Daniels)",female,25.0,1,2,113781,151.55,C22 C26,S,,,,"Montreal,
  PQ / Chesterville, ON"
7 1,1,"Anderson, Mr. Harry",male,48.0,0,0,19952,26.55,E12,S,3,,,"New York, NY"
8 1,1,"Andrews, Miss. Kornelia Theodosia",female,63.0,1,0,13502,77.9583,D7,S,10,,,"Hudson, NY"
9 1,0,"Andrews, Mr. Thomas Jr",male,39.0,0,0,112050,0.0,A36,S,,,,"Belfast, NI"
10 1,1,"Appleton, Mrs. Edward Dale (Charlotte Lamson)",female,53.0,2,0,11769,51.4792,C101,S,D,,,"Bayside,
  Queens, NY"
11 1,0,"Artagaveytia, Mr. Ramon",male,71.0,0,0,PC 17609,49.5042,,C,,22.0,"Montevideo, Uruguay"
12 1,0,"Astor, Col. John Jacob",male,47.0,1,0,PC 17757,227.525,C62 C64,C,,124.0,"New York, NY"
13 1,1,"Astor, Mrs. John Jacob (Madeleine Talmadge Force)",female,18.0,1,0,PC 17757,227.525,C62 C64,C,4,,,"New
  York, NY"
14 1,1,"Aubart, Mme. Leontine Pauline",female,24.0,0,0,PC 17477,69.3,B35,C,9,,,"Paris, France"
15 1,1,"Barber, Miss. Ellen 'Nellie'",female,26.0,0,0,19877,78.85,,S,6,,,"
16 1,1,"Barkworth, Mr. Algernon Henry Wilson",male,80.0,0,0,27042,30.0,A23,S,B,,,"Hessle, Yorks"
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VALUE ENCODINGS: PRIMITIVE TYPES

1. NUMBERS & STRINGS

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- Date
- Time
- Geolocations



surprisingly subtle and MESSY

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2. COMMON SPECIAL CASES 'PRIMITIVE TYPES'

- Date
- Time
- Geolocations

} surprisingly subtle and MESSY

DATE 1	DATE 2	LOC 1	LOC 2
25th April 2019	25/04/2019	North Holland	NH
25th April 2019	04/25/2019	Friesland	FR
25th April 2019	25.04.2019	Fryslân	FR

PRIMITIVE TYPES **SEMANTICS**

CATEGORICAL

- **NOMINAL**

‘Gender’, ‘Embarked’, ‘Cabin’

- **ORDINAL**

‘Pclass’, ‘Survived’

PRIMITIVE TYPES **SEMANTICS**

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- **NOMINAL**

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'Pclass'

NUMERIC DATA CAN BE CATEGORICAL
ORDERED BUT NOT QUANTITATIVE

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NUMERIC DATA CAN BE CATEGORICAL
ORDERED BUT NOT QUANTITATIVE

QUANTITATIVE

- **AMOUNTS & MEASURES**

negative and positive

arithmetic can be performed

e.g. differences and ratios

integers VS rational numbers

RECAP: ASSESSING STRUCTURE

1. COARSE STRUCTURE

- Identify 'data shape': records/fields and delimiters
- Value encodings: categorical VS quantitative

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- Value encodings: categorical VS quantitative

2. BASIC TOOL

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- Applications

3. SIMPLE DATA TRANSFORMATION

- Edit-in-place, character-by-character or Find/Replace - **IMPORTANT: file versioning!**
- Scripted transformations - **IMPORTANT: save scripts, inputs and outputs!**

WHAT ABOUT NOT RECTANGULAR DATA



**Big Data
Borat**

@BigDataBorat



Following

In Data Science, 80% of time spent prepare data, 20% of time spent complain about need for prepare data.



**NATURAL LANGUAGE
NESTED DATA**

WHAT ABOUT NOT RECTANGULAR DATA



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**NATURAL LANGUAGE
NESTED DATA**

```
{
  "created_at": "Thu Sep 26 14:30:00 +0000 2023",
  "id": 1707082345678901234,
  "id_str": "1707082345678901234",
  "text": "In Data Science, 80% of time spent prepare data,
20% of time spent complain about need for prepare data.",
  "user": {
    "id": 987654321,
    "id_str": "987654321",
    "name": "Big Data Borat",
    "screen_name": "BigDataBorat",
    "location": "Kazakhstan",
    "description": "Making glorious data jokes! High five! 🙌",
    "followers_count": 3000,
    "friends_count": 100,
    "statuses_count": 500,
    "verified": false,
    "created_at": "Mon Apr 20 10:20:00 +0000 2015"
  },
  "retweet_count": 100,
  "favorite_count": 500,
  "lang": "en"
}
```


PLAYING WITH NATURAL LANGUAGE

1. NATURAL LANGUAGE PROCESSING (NLP) 🥶🏂🧊🏄

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(e.g. 'IBM' —> 'International Business Machines')

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'My Windows computer crashed yet again!'

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AND MANY OTHERS

ASSESSING GRANULARITY

level of details in the data structure

ASSESSING GRANULARITY

level of details in the data structure

1. WHAT DOES EACH TITANIC RECORD REPRESENTS?
2. WHAT IS THE GRANULARITY OF EACH RECORD?
3. WHAT IS THE PRIMARY KEY OF THE DATASET?

A pug dog is lying on a bed, wrapped in a brown, textured blanket. The dog's face is visible, looking directly at the camera. The background is a soft-focus view of a bed with white pillows and a light-colored headboard. A white speech bubble with a tail pointing towards the dog contains the text "TIME FOR A BREAK" in bold, black, uppercase letters.

**TIME FOR
A BREAK**

A Primary Key is a designated attribute (column) that can reliably *identify* and *distinguish* between each individual record ...

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- ✿ e.g. identifier, multiple attributes (composite key)
- ✿ each value occurs **at most once** in the key column(s)
- ✿ the semantics (meaning) of the key determines granularity

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BE CAREFUL!

GOOFY KEY? GOOFY DATA

REAL DATA MAY HAVE NOISE & DUPLICATED KEYS

Join at menti.com | use code 2844 5236



ASSESSING GRANULARITY

level of details in the data structure

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Each record captures attributes and characteristics of a passenger

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One record per passenger, granularity on the individual level

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Passenger ID, composite key, might have to make a new one!

A Primary Key is a *unique identifier* within its table.

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A Foreign Key is a *reference* on one table to a primary key in another table.

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Primary keys **enforce uniqueness** within their table, **ensuring each record is identifiable**.

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THIS IS HOW YOU JOIN TABLES!

Customers		
CustomerNo	FirstName	LastName
1	Sally	Thompson
2	Sally	Henderson
3	Harry	Henderson
4	Sandra	Wellington
6	Arthur	Paulson
7	Derry	Beckham
8	Maude	Jones

Primary key

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4	Sandra	Wellington
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Primary key

Orders					
OrderNo	EmployeeNo	CustomerNo	Supplier	Price	Item
1	1	42	Harrison	\$235	desk
2	4	1	Ford	\$234	chair
3	1	68	Harrison	\$415	table
4	2	4	Ford	\$350	lamp
5	3	6	Ford	\$234	chair
6	2	112	Ford	\$350	lamp
7	2	8	Harrison	\$235	desk

Foreign key

GRANULARITY: A CHECKLIST

- 1. WHAT DOES EACH RECORD REPRESENTS?**
- 2. DO ALL RECORDS CAPTURE GRANULARITY AT THE SAME LEVEL?**
- 3. WHAT ALTERNATIVE INTERPRETATIONS OF THE RECORD ARE THERE?**

e.g. is this a list of passengers? a list of sold tickets?

- 4. WHAT KIND OF AGGREGATION IS POSSIBLE OR DESIRABLE?**

e.g. from individual to group level? From individual events to total across time and space?

Going up the hierarchy (town/regions/country)?

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DATA MODIFICATION —> GRANULARITY MODIFICATION

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1. THE LEVEL OF GRANULARITY CHANGES IF YOU MAKE MODIFICATIONS

2. IDENTIFYING PRIMARY KEYS

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3. FOREIGN KEYS ARE POINTERS TO INDIVIDUAL ROWS IN ANOTHER DATASETS

Want to join tables? Join foreign keys to primary keys

ASSESSING FAITHFULNESS

level of trust and factual consistency in the data

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1. MISSING VALUES

- Missing not at random (MNAR)
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FAITHFULNESS CAN ONLY BE EVALUATED IN CONTEXT

Missing Completely at Random (MCAR)

The missing data has no relationship with any other variables or the value that is missing itself. Completely random and doesn't depend on anything observable.

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KEY IDEA

THE MISSING VALUES HAVE NO PATTERN

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The missing data has a systematic relationship with some of the observed data, but not with the missing data itself.

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**MISSINGNESS DEPENDS ON OTHER OBSERVED
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KEY IDEA

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An outlier is a data point that is significantly different from other observations. Outliers can occur due to variability, errors or extreme TRUE observations.

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KEY IDEA

A VALUE THAT STANDS OUT

STRUCTURE	GRANULARITY	FAITHFULNESS	TEMPORALITY	SCOPE
	EXPLANATION	EXAMPLE		
MCAR (Missing Completely at Random)	The missing values have no pattern	Information missing because of system error		
MAR (Missing at Random)	Missingness depends on other variables	Younger people less likely to report income		
MNAR (Missing Not at Random)	Missingness depends on the missing value itself	Older people less likely to report age (because of eligibility to aid)		
Outliers	Data points significantly different from the rest	A person aged 120, among 30-70 year olds		

WHAT TO DO WITH MISSING VALUES AND OUTLIERS?

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BE CAREFUL!

WHAT TO DO DEPENDS ON YOUR GOAL AND THE CONTEXT

FAITHFULNESS: A CHECKLIST

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- It is what it is
- Human errors
- Data is confidential (e.g. address, BSN)

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3. CAN YOU DO SOMETHING ABOUT IT?

- That depends!

ASSESSING TEMPORALITY (quick note)

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TIME IS REAAAALLLYYYY TRICKY!

Month Recording	Day	Hours of sun	Time of sunrise
July 2024	10.07.2024	16:28	04:43
July 2024	25.07.2024	15:51	05:07
August 2024	07.08.2024	13:11	06:27
August 2024	13.08.2024	12:47	06:37

ASSESSING TEMPORALITY (quick note)

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Time of data entry VS time of recorded phenomenon

Periodicity (e.g. days of the week, months of the year)

Non-uniform (# days in a month, # days in a year)

Time zones

Clocks are not always correct

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BE CAREFUL!
ASSESS TIMESTAMPS

ASSESSING SCOPE (quick note)

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APPLICATION SPECIFIC & SUBJECTIVE!

TO CONCLUDE

1. WHAT IS DATA WRANGLING?

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4. WHAT DECISIONS DID WE MAKE ALONG THE WAY AND WHY?

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3. WHAT ARE THE STEPS OF DATA WRANGLING?

4. WHAT DECISIONS DID WE MAKE ALONG THE WAY AND WHY?

5. HOW CAN DATA WRANGLING INFLUENCE DATA ANALYSIS?

